

Question number	Answer	Additional guidance	Mark
7(a)	An answer that includes the following points: • active immunotherapy {stimulates the immune system / causes an immune response / uses (cancer) antigens} (1) • so memory cells generated (1)	ACCEPT description of what is stimulated e.g. B cells, antibody production	(2)

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(b)(i)	An explanation that includes the following points: - because a mutation causes a change in the {DNA / base} sequence (1) - and this can result in a {change in the / new} {amino acid sequence / primary structure} (1)	ACCEPT change in genetic code change in one base ACCEPT (poly) peptide change in one amino acid	(2)

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(b)(ii)	An explanation that includes four of the following points: - the mRNA results in the production of the neoantigens (1) - the body will recognise these neoantigens as foreign (1) - therefore macrophages will {engulf the neoantigens / express neoantigens on surface} (1) - and macrophages present the neoantigens to {T helper / CD4} cells (1) - which will then release cytokines to stimulate {an immune response / humoral response / cell-mediated immunity / B cells / T killer cells} (1)	ACCEPT mRNA translated (into neoantigens) ACCEPT present on surface / express on MHC ACCEPT macrophages become APC {to T_h cells / and T helper cells bind to {macrophage / neoantigen / MHC}} ACCEPT chemicals for cytokines	(4)

Question number	Answer	Additional guidance	Mark
7(c)	An explanation that includes the following points: - because the antibodies will {bind to the neoantigens / form an antigen antibody complex} (1) - and therefore the cancer cells will be destroyed faster by the {macrophages / phagocytes} (1) OR - antibodies binding cancer cells together_ (1) - and therefore the cancer cells will be {destroyed faster by the macrophages / prevented from spreading} (1)	ACCEPT opsonisation (of cancer cells) ACCEPT engulfed by macrophages faster ACCEPT agglutination (of cancer cells) ACCEPT engulfed by {macrophages / phagocytes} faster	(2)

Question number	Answer	Additional guidance	Mark
7(d)	An explanation that includes the following points: - because large number (T helper cells) are needed to activate {B cells / T killer cells} (1) - because genetically identical cells will not be rejected (1)	IGNORE pathogens throughout ACCEPT large number (T killer cells) are needed to destroy cancer cells effects of {HI / CMI} e.g. more antibody produced ACCEPT need to be specific to the (cancer cell) antigens	(2)

Question number	Answer	Additional guidance	Mark
7(e)	An answer that includes two of the following points: - must not be {pathogenic / cause disease / cause cancer} (1) - must {be specific to / target} the {cancer cells / neoantigens} (1) - must be able to destroy the cancer cells (1)	ACCEPT can bind to cancer cells / can enter cancer cells / only affect cancer cells / do not affect healthy cells	(2)

